



Ata Kalirad, PhD

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Summary

- Senior postdoctoral researcher in evolutionary biology.
- Currently focused on understanding the importance of **phenotypic plasticity** in the ecological and evolutionary contexts, specifically how it determines the composition of ecological communities.
- Proven record of combining computational and experimental approaches to address complex biological questions.

Technical Skills

- **Programming Languages:** Python (+10 years), PyMC (+3 years), R, and Mathematica
- **Software and Tools:** L^AT_EX, PGF/TikZ, Git/GitHub, Pandas, NumPy, Scikit-Learn, OpenCV, Matplotlib, Seaborn, NetworkX, MS Office
- **Laboratory Skills:** Experience using nematodes, specifically strains in genus *Pristionchus* as a model organism (+4 years), utilizing computer vision methods to analyze microscopic images ($\approx$ 1 year)
- **Mathematical and Statistical Skills:** Extensive experience in mathematical modeling, especially individual-based models; extensive knowledge of statistical methods, especially Bayesian approaches

Language Skills

English: Proficient | **German:** B1 | **Farsi:** Mother tongue

Experience

Senior Postdoctoral Fellow, Max-Planck-Institut für Biologie Tübingen – Germany March 2020 – Present

- Developed Bayesian models to infer accurate estimates of life-history traits based on laboratory data.
- Developed agent-based models, combined with probabilistic programming, to simulate species interaction in communities and meta-communities.
- Developed software to acquire and analyze large life-history data using computer vision for the loopbio high-throughput imaging system.
- Developed stage-structured models to investigate ecological implications of phenotypic plasticity.
- Supervised PhD students to use appropriate statistical and visualization approaches to analyze biological data.

Postdoctoral Fellow, Institute for Research in Fundamental Sciences – Tehran, Iran Oct 2017 – Feb 2020

- Led and collaborated on a project to develop a game-theoretical framework to understand the choice of the ATP-producing pathways in yeast.
- Led a collaborative project, involving physicist and computer scientists, to develop a stochastic model of cell division and proliferation that explains the role of noise in the emergence of spatial and temporal organization in biological systems, particularly in cell differentiation.

Education

University of Houston, PhD in Biology Aug 2011 – Dec 2016

- **Thesis:** *A Test of the Snowball Model* under the supervision of Ricardo B. R. Azevedo
- GPA: 3.818/4.0

University of Tehran, Iran, BS in Cell & Molecular Biology July 2007 – July 2011

Projects

Selected Projects (For a complete list, see github):

PriPOP

github repo

- An agent-based model, written as an OOP, to simulate the population dynamics of a nematode using experimental data in a single or multiple populations.
- Tools Used: Python, PyMC, Jupyter notebook

Speciation Model

github repo

- Models of sequence evolution based on the Russian roulette and RNA folding models to simulate the accumulation of genetic incompatibilities in diverging populations in allopatry.
- Tools Used: Python, ViennaRNA Package, Jupyter notebook

Making ATP fast and slow

github repo

- A repository of an individual-based model, based on the *Gillespie algorithm*, to understand the switch between ATP-producing paths in a cell.
- Tools Used: Python, Jupyter notebook

Conferences and Events

- **Mathematical Models in Ecology and Evolution**, July 2024, University of Vienna, Wien, Austria
- **4th International *Pristionchus* Meeting**, September 2023, Tübingen, Germany
- **Plasticity across scales: from molecules to phenotypes**, October 2022, EMBL Heidelberg, Germany
- **The ICTS-ICTP Winter School on Quantitative Systems Biology**, December 2019, Bengaluru, India
- **International Biology Olympiad**, July 2018, Tehran, Iran (Part of the scientific committee)
- **Evolution Meeting**, June 2016, Austin, TX, USA

Teaching and Mentoring Experiences

- 2024 - Present: Mentoring undergraduate students to conduct rigorous scientific research in biology, part of a non-profit project to provide high-quality mentoring for talented Iranian students biological sciences.
- Spring semester 2019: **Species and speciation**, Graduate-level course, University of Tehran, School of Biology (Department of Zoological Sciences)

Selected Publications

15 publications in international peer-reviewed scientific journals, Complete list: Google Scholar profile

The role of plasticity and stochasticity in coexistence

2024

Ata Kalirad, Ralf J Sommer

Ecology Letters, 27, e14370, DOI: 10.1111/ele.14370

Genetic drift promotes and recombination hinders speciation on holey fitness landscapes

Ata Kalirad, Christina L Burch, Ricardo BR Azevedo

PLoS Genetics, 20(1): e1011126, DOI: 10.1371/journal.pgen.1011126

Spatial and temporal heterogeneity alter the cost of plasticity in *Pristionchus pacificus*

Ata Kalirad, Ralf J Sommer,

PLoS Computational Biology, 20(1): e1011823, DOI: 10.1371/journal.pcbi.1011823

Spiraling Complexity: A Test of the Snowball Effect in a Computational Model of RNA Folding (*Highlighted article*)

2017

Ata Kalirad, Ricardo B R Azevedo,

Genetics, Volume 206, Issue 1, 1 May 2017, Pages 377–388, DOI: 10.1534/genetics.116.196030